

Program 9:

Organize Kubernetes Pods using Labels and Namespace

Project Structure:

program_9/

—>namespaces.yaml

—> pods.yaml

Procedure:

Create a folder (ex: program_9)

Step 1: Start Minikube

minikube start

Check the status of Minikube

minikube status

```
31 apiVersion: v1
```

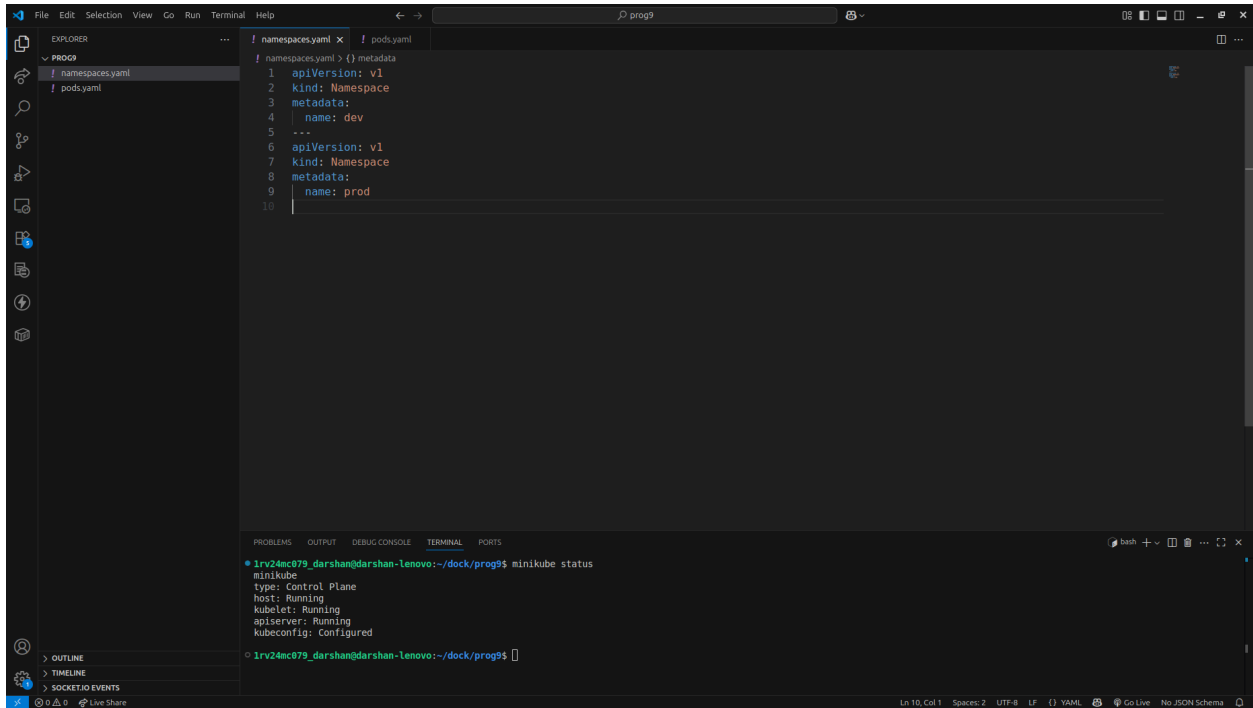
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
```

```
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ minikube status  
minikube  
type: Control Plane  
host: Running  
kubelet: Running  
apiserver: Running  
kubecfg: Configured
```

```
○ 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ █
```

Step 2: Create namespaces.yaml file with the following code

nano namespaces.yaml



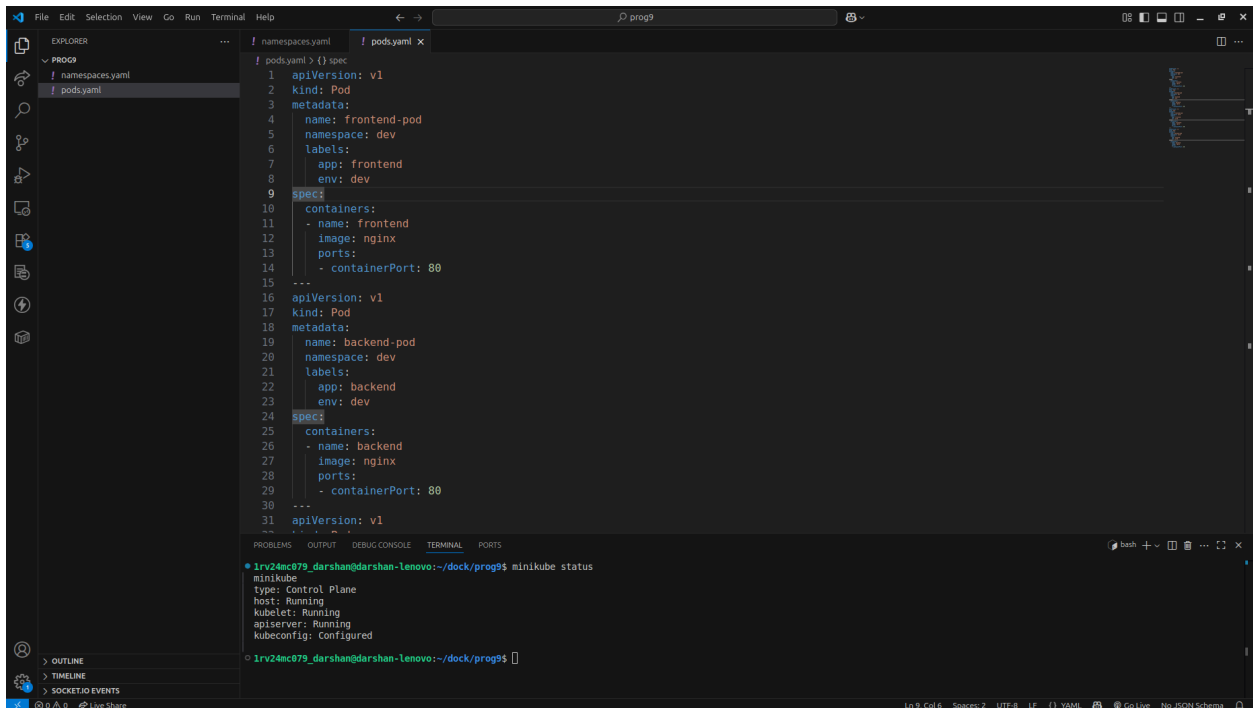
The screenshot shows the Visual Studio Code editor with two files open: `namespaces.yaml` and `pod.yaml`. The `namespaces.yaml` file contains the following YAML code:

```
1 apiVersion: v1
2 kind: Namespace
3 metadata:
4   name: dev
5 ---
6 apiVersion: v1
7 kind: Namespace
8 metadata:
9   name: prod
```

The terminal at the bottom shows the command `minikube status` being executed, with the following output:

```
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create `pod.yaml` file with the following code
`nano pod.yaml`



The screenshot shows the Visual Studio Code editor with two files open: `namespaces.yaml` and `pod.yaml`. The `pod.yaml` file contains the following YAML code:

```
1 apiVersion: v1
2 kind: Pod
3 metadata:
4   name: frontend-pod
5   namespace: dev
6   labels:
7     app: frontend
8     env: dev
9 spec:
10   containers:
11     - name: frontend
12       image: nginx
13       ports:
14         - containerPort: 80
15 ---
16 apiVersion: v1
17 kind: Pod
18 metadata:
19   name: backend-pod
20   namespace: dev
21   labels:
22     app: backend
23     env: dev
24 spec:
25   containers:
26     - name: backend
27       image: nginx
28       ports:
29         - containerPort: 80
30 ---
31 apiVersion: v1
```

The terminal at the bottom shows the command `minikube status` being executed, with the following output:

```
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 4: Apply the files
`kubectl apply -f namespaces.yaml`

kubectl apply -f pods.yaml

```
31 apiVersion: v1
22

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
○ 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$
```

Step 5: Verify the pods with namespaces

kubectl get pods -n dev

kubectl get pods -n prod

```
31 apiVersion: v1
22

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-65d889d5b8-2p7jg   1/1     Running   0           68s
nginx-deployment-65d889d5b8-mnhp9   1/1     Running   0           68s
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           6m15s
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get deployments.apps
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2           6m17s
○ 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$
```

Step 6: Get all of the pods

docker get pods -A

Step 7: Verify pods with labels

kubectl get pods -n dev -l app=frontend

kubectl get pods -n dev -l app=backend

kubectl get pods -n prod -l app=frontend

kubectl get pods -n prod -l app=backend

```
31 apiVersion: v1
22

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get pods -n dev -l app=frontend
NAME          READY   STATUS             RESTARTS   AGE
frontend-pod  0/1     ContainerCreating   0           51s
• 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$ kubectl get pods -n dev -l app=backend
NAME          READY   STATUS             RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0           57s
○ 1rv24mc079_darshan@darshan-lenovo:~/dock/prog9$
```